

Which of the following is FALSE about $\lim_{x \rightarrow \infty} \frac{x + \cos x}{2x + 1}$?

- ☒ It equals $\lim_{x \rightarrow \infty} \frac{1 - \sin x}{2}$ by l'Hospital's rule. ✓
- ☐ It exists.
- ☐ It equals $\frac{1}{2}$.
- ☐ It is indeterminate of type $\frac{\infty}{\infty}$.

2. (3 Points)

Given that $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ and $\lim_{x \rightarrow a} p(x) = \lim_{x \rightarrow a} q(x) = \infty$, which of the following limits is NOT an indeterminate form?

- ☒ $\lim_{x \rightarrow a} [p(x)]^{q(x)}$ ✓
- ☐ $\lim_{x \rightarrow a} [1 + p(x)]^{f(x)}$
- ☐ $\lim_{x \rightarrow a} [1 + f(x)]^{p(x)}$
- ☐ $\lim_{x \rightarrow a} [f(x)]^{q(x)}$

3. (3 Points)

$\int 2 \cos(5x) \cos(2x) \, dx = \underline{\hspace{2cm}} + C$

- ☒ $\frac{1}{7} \sin(7x) + \frac{1}{3} \sin(3x)$ ✓
- ☐ $\frac{1}{14} \cos(7x) + \frac{1}{6} \sin(3x)$
- ☐ $-\frac{1}{7} \cos(7x) - \frac{1}{3} \cos(3x)$
- ☐ $-\frac{1}{7} \sin(7x) + \frac{1}{3} \sin(3x)$

4. (3 Points)

$\int \frac{dx}{x^2 \sqrt{x^2 - 1}} = \underline{\hspace{2cm}}$

- ☒ $\int \cos \theta \, d\theta$ where $\theta = \operatorname{arccsc} x$ ✓
- ☐ $\int \cos \theta \, d\theta$ where $\theta = \arcsin x$
- ☐ $\int \frac{d\theta}{\tan^2 \theta}$ where $\theta = \operatorname{arcsec} x$
- ☐ $\int \frac{d\theta}{\sin^2 \theta}$ where $\theta = \arcsin x$

6. (3 Points)

$\int \frac{dx}{x^{\frac{1}{2}} + x^{\frac{3}{2}}} = \underline{\hspace{2cm}} + C$

- ☒ $2 \arctan \sqrt{x}$ ✓
- ☐ $\arctan \sqrt{x}$
- ☐ $\ln \left| \frac{\sqrt{x} - 1}{\sqrt{x} + 1} \right|$
- ☐ $\frac{1}{2} \ln \left| \frac{\sqrt{x} - 1}{\sqrt{x} + 1} \right|$

7. (3 Points)

Which of the following statements is TRUE?

- ☒ $\int_{-\infty}^{\infty} \frac{dx}{x^2 + 1} = \lim_{t \rightarrow \infty} \int_{-t}^t \frac{dx}{x^2 + 1}$ ✓
- ☐ $\int_{-\infty}^{\infty} \sin x \, dx = \lim_{t \rightarrow \infty} \int_{-t}^t \sin x \, dx$
- ☐ $\int_{-\infty}^{\infty} 2 \, dx = \lim_{t \rightarrow \infty} \int_{-t}^t 2 \, dx$
- ☐ $\int_{-\infty}^{\infty} x^2 \, dx = \lim_{t \rightarrow \infty} \int_{-t}^t x^2 \, dx$

8. (3 Points)

The sequence of partial sums of $\sum_{n=0}^{\infty} \frac{1}{n}$ is _____.

- ☒ monotonic ✓
- ☐ bounded
- ☐ convergent
- ☐ decreasing

9. (3 Points)

Which of the following statements is TRUE?

- ☐ If $\int_1^{\infty} f(x) \, dx$ is convergent, then $\sum_{n=1}^{\infty} f(n)$ is convergent.
- ☐ If $\sum_{n=1}^{\infty} f(n)$ is convergent, then $\int_1^{\infty} f(x) \, dx$ is convergent.
- ☒ $\int_1^{\infty} \frac{1}{x^{\frac{1}{3}}} \, dx$ is convergent. ✓
- ☐ $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{6}}}$ is divergent.

10. (3 Points)

$\lim_{n \rightarrow \infty} \frac{100^n}{n!} = \underline{\hspace{2cm}}$

- ☒ 0 ✓
- ☐ e^{100}
- ☐ e^{-100}
- ☐ 1

11. (3 Points)

Which of the following holds for the series $\sum_{n=0}^{\infty} \frac{(-4)^n}{(2n)!}$?

- ☒ It converges to the sum $\cos 2$ ✓
- ☐ It converges to the sum $\cosh 2$
- ☐ It converges to the sum $\frac{1}{2} \sin 2$
- ☐ It converges to the sum $\frac{1}{2} \sinh 2$

12. (4 Points)

Find the first four terms of the Maclaurin series for $\cos(x + \pi/2)$.

- ☒ $-x + \frac{x^3}{6}$ ✓
- ☐ $-x + x^3$
- ☐ $1 - \frac{x^2}{2}$
- ☐ $1 - x^2$

What is the radius of convergence of the power series $\sum_{n=0}^{\infty} \left(\frac{n-2}{n} \right)^{n^2} x^n$?

- ☒ e^2 ✓
- ☐ e^{-2}
- ☐ 0
- ☐ ∞

14. (3 Points)

If $f'(x) = \frac{1}{1-x^3}$, which of the following is a power series representation of f ?

- ☐ $C + \sum_{n=0}^{\infty} \frac{x^{3n+1}}{3n+1}, \quad |x| < 1$ ✓
- ☐ $C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{3n+1}, \quad |x| < 1$
- ☐ $C + \frac{1}{3} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}, \quad |x| < 1$
- ☐ $C + \frac{1}{3} \sum_{n=1}^{\infty} \frac{(-1)^n x^{n+1}}{n+1}, \quad |x| < 1$

If $f(x) = \sum_{n=2}^{\infty} \frac{(x-2)^n}{n^3}$, then $f^{(10)}(2) = \underline{\hspace{2cm}}$.

- ☒ $\frac{9!}{100}$ ✓
- ☐ $\frac{10!}{100}$
- ☐ $\frac{1}{1000}$
- ☐ $\frac{2^{10}}{1000}$

16. (3 Points)

Which of the following holds for the series $\sum_{n=0}^{\infty} \left(\frac{1}{n} \right) 2^n$?

- ☒ It diverges ✓
- ☐ It converges to the sum $\sqrt{3}$
- ☐ It converges to the sum $-\sqrt{3}$
- ☐ It converges to the sum $\left(\frac{1}{2} \right)$

The region enclosed by the curves $y = x$ and $y = \sqrt{4x}$ is rotated about the x -axis. Find the volume of the resulting solid.

- ☒ $\frac{32\pi}{3}$ ✓
- ☐ 32π
- ☐ $\frac{4\pi}{3}$
- ☐ 4π

18. (4 Points)

Which integral represents the length of the arc of the curve $y = x^2 - x + 2$ from the point $P(1, 2)$ to the point $Q(3, 8)$?

- ☒ $\int_1^3 \sqrt{4x^2 - 4x + 2} \, dx$ ✓
- ☐ $\int_2^8 \sqrt{4x^2 - 4x + 2} \, dx$
- ☐ $\int_1^3 \sqrt{4x^2 + 4x + 2} \, dx$
- ☐ $\int_2^8 \sqrt{4x^2 + 4x + 2} \, dx$

19. (3 Points)

Which of the following represents the area of the surface obtained by rotating the curve $y = \sqrt{1 + e^x}$, $0 \leq x \leq 1$ about the x -axis?

- ☒ $\pi \int_0^1 (e^x + 2) \, dx$ ✓
- ☐ $2\pi \int_2^{1+e} (e^x + 2) \, dx$
- ☐ $\pi \int_0^1 \frac{e^x + 2}{\sqrt{1 + e^x}} \, dx$
- ☐ $\pi \int_0^1 \frac{xe^x + 2x}{\sqrt{1 + e^x}} \, dx$

20. (4 Points)

Let $(\bar{x}, \bar{y})$ denote the centroid of the region enclosed by the curves $y = x^3$, $y = 0$, and $x = 3$. Find $\bar{x}$.

- ☒ $\frac{12}{5}$ ✓
- ☐ $\frac{3}{5}$
- ☐ $\frac{54}{7}$
- ☐ $\frac{27}{7}$

21. (4 Points)

Find the point on the curve $x = t^2 + 1$, $y = t^2 - 4t - 3$ where the tangent line is parallel to the line $y = 2x$.

- ☐ $(-1, 7)$
- ☐ $(-1, -6)$
- ☒ $(5, 9)$ ✓
- ☐ $(5, 7)$

22. (3 Points)

Suppose that $f'(2) = g'(2) = 1$. Which of the following is true about the tangent line to the curve $x = t + f(t)$, $y = t - g(t)$ at the point where $t = 2$?

- ☐ It is vertical.
- ☒ It is horizontal. ✓
- ☐ It has slope 1.
- ☐ It has slope -1 .

23. (4 Points)

For what values of t is the curve $x = t^2 - 1$, $y = t^3$ concave upward?

- ☒ $t \in (0, \infty)$ ✓
- ☐ $t \in (-1, 1)$
- ☐ $t \in (-\infty, 0)$
- ☐ $t \in \mathbb{R} \setminus \{0\}$

24. (4 Points)

Find the area enclosed by the curve $x = 4t$, $y = 2t - t^2$ and the x -axis.

- ☐ 4
- ☐ 1
- ☐ $\frac{5}{3}$
- ☒ $\frac{16}{3}$ ✓

25. (3 Points)

Which of the following integrals represents the length of the curve $x = t - 2 \sin t$, $y = 2 \cos t$, $0 \leq t \leq \pi$?

- ☒ $\int_0^{\pi} \sqrt{5 - 4 \cos t} \, dt$ ✓
- ☐ $\int_0^{\pi} \sqrt{5 + 4 \sin t} \, dt$
- ☐ $\int_0^{\pi} \sqrt{2 - 4 \sin t} \, dt$
- ☐ $\int_0^{\pi} \sqrt{2 + 4 \cos t} \, dt$

26. (3 Points)

Which of the following is a polar representation of the Cartesian point $(-\sqrt{3}, -1)$?

- ☒ $(2, 7\pi/6)$ ✓
- ☐ $(2, 5\pi/6)$
- ☐ $(2, 4\pi/3)$
- ☐ $(2, 2\pi/3)$

27. (3 Points)

Which polar curve is symmetric about the polar axis?

- ☒ $r = \frac{\sin \theta}{\theta}$ ✓
- ☐ $r = 2 \sin \theta$
- ☐ $r = 2 \sin \theta + 3 \cos \theta$
- ☐ $r = 1 + \sin \theta$

28. (4 Points)

Find the slope of the tangent line to the curve $r = \sin(3\theta)$ at the point where $\theta = \frac{\pi}{4}$.

- ☒ $\frac{1}{2}$ ✓
- ☐ $\frac{1}{\sqrt{2}}$
- ☐ $\frac{3}{2}$
- ☐ $\sqrt{2}$

29. (4 Points)

Find the area of the region enclosed by the polar curve $r = \sqrt{2 \sin \theta}$.

- ☐ 1
- ☐ $\frac{1}{2}$
- ☒ 2 ✓
- ☐ 0

30. (4 Points)

Which of the following represents the exact length of one loop of the four-leaved rose $r = \cos(2\theta)$?

- ☒ $2 \int_0^{\frac{\pi}{4}} \sqrt{1 + 3 \sin^2(2\theta)} \, d\theta$ ✓
- ☐ $2 \int_0^{\frac{\pi}{4}} \sqrt{5} \, d\theta$
- ☐ $\int_0^{\frac{\pi}{4}} \cos^2(2\theta) \, d\theta$
- ☐ $2 \int_0^{\frac{\pi}{4}} \sqrt{\cos(4\theta)} \, d\theta$